Decoding/Inversion Ambiguity Under Noise: A Formal Analysis

Quantum Communication Theory

Week 2: Noisy Quantum Channels

1 Introduction

This document provides a rigorous analysis of decoding and inversion ambiguity in quantum communication systems under noise. We formalize the fundamental ambiguity that arises when attempting to invert noisy quantum channels and characterize its implications for reliable communication.

2 Mathematical Framework

2.1 Inversion Problem Formulation

Consider a quantum communication system with:

- Input alphabet $\mathcal{X}$ with probability distribution $p(x)$
- Input quantum states $\{\rho_x\}_{x\in\mathcal{X}}$
- Noisy quantum channel $\mathcal{N} : \mathcal{L}(\mathcal{H}_A) \to \mathcal{L}(\mathcal{H}_B)$
- Measurement $\mathcal{M} : \mathcal{L}(\mathcal{H}_B) \to \mathcal{Y}$ (classical outcomes)

The decoding/inversion problem: Given measurement outcome $y \in \mathcal{Y}$, estimate the original input $x \in \mathcal{X}$.

2.2 Mathematical Statement of Ambiguity

Definition 1 (Decoding Ambiguity)**.** *The decoding problem has ambiguity if the map*

$$f : \mathcal{X} \to \mathcal{Y}, \quad f(x) = \mathbb{P}(y|x)$$

is not injective, where $\mathbb{P}(y|x) = Tr[\Pi_y \mathcal{N}(\rho_x)]$ *and* $\{\Pi_y\}$ *are POVM elements.*

Definition 2 (Inversion Ambiguity)**.** *The inversion problem has ambiguity if the map*

$$g : \mathcal{D}(\mathcal{H}_A) \to \mathcal{D}(\mathcal{H}_B), \quad g(\rho) = \mathcal{N}(\rho)$$

is not injective, where $\mathcal{D}(\mathcal{H})$ *denotes density operators on* $\mathcal{H}$*.*

3 Sources of Ambiguity

3.1 Non-Injectivity of Noisy Channels

Theorem 1 (Channel Non-Injectivity)**.** *For any non-unitary quantum channel* $\mathcal{N}$ *(i.e., any channel with noise), there exist distinct input states* $\rho_1 \neq \rho_2$ *such that:*

$$\mathcal{N}(\rho_1) = \mathcal{N}(\rho_2)$$

Proof. Let $\mathcal{N}$ be a non-unitary CPTP map. Then $\mathcal{N}$ has non-trivial kernel or collapses states. Consider:

- For amplitude damping: $\mathcal{A}_\gamma(|1\rangle\langle 1|) = \mathcal{A}_\gamma(\rho_{\text{mixed}})$ for certain mixed states
- For depolarizing: $\mathcal{D}_p(\rho) = \mathcal{D}_p(\rho')$ for any ρ, ρ' when $p = \frac{d}{d+1}$
- In general: Any channel with non-zero noise parameter has collapsing pairs

□

3.2 Measurement Ambiguity

Theorem 2 (POVM Ambiguity)**.** *For any non-projective POVM measurement* $\{\Pi_y\}$*, there exist distinct states* $\rho_1 \neq \rho_2$ *such that:*

$$\mathbb{P}(y|\rho_1) = \mathbb{P}(y|\rho_2) \quad \forall y \in \mathcal{Y}$$

4 Formal Characterization of Ambiguity

4.1 Ambiguity Sets

Definition 3 (Ambiguity Set)**.** *For a given output $\sigma = \mathcal{N}(\rho)$, the ambiguity set is:*

$$\mathcal{A}(\sigma) = \{\rho' \in \mathcal{D}(\mathcal{H}_A) : \mathcal{N}(\rho') = \sigma\}$$

For measurement outcome y, the decoding ambiguity set is:

$$\mathcal{A}_D(y) = \{x \in \mathcal{X} : \arg\max_{x'} \mathbb{P}(x'|y) \text{ is not unique}$$

4.2 Ambiguity Measure

Definition 4 (Ambiguity Measure)**.** *The ambiguity of channel $\mathcal{N}$ can be quantified by:*

$$\mathcal{A}(\mathcal{N}) = \sup_{\sigma \in Image(\mathcal{N})} \frac{|\mathcal{A}(\sigma)|}{\dim(\mathcal{H}_A)}$$

where $|\mathcal{A}(\sigma)|$ measures the size of the ambiguity set.

Definition 5 (Decoding Error Basin)**.** *For a decoding rule $\hat{x} : \mathcal{Y} \to \mathcal{X}$, the error basin for input x is:*

$$\mathcal{B}_\epsilon(x) = \{y \in \mathcal{Y} : \hat{x}(y) \neq x \text{ but } \mathbb{P}(y|x) \geq \epsilon\}$$

5 Analytical Results for Specific Channels

5.1 Depolarizing Channel Ambiguity

Theorem 3 (Depolarizing Ambiguity)**.** *For the depolarizing channel $\mathcal{D}_p(\rho) = (1-p)\rho + p\frac{I}{d}$:*

1. *For $p \geq \frac{d}{d+1}$, $\mathcal{D}_p$ maps all inputs to the maximally mixed state*

2. *For $0 < p < \frac{d}{d+1}$, the ambiguity set for output σ is:*

$$\mathcal{A}(\sigma) = \left\{\rho : \rho = \frac{\sigma - \frac{p}{d}I}{1-p} \text{ is positive}\right\}$$

which is non-empty only when σ is sufficiently close to $\frac{I}{d}$

5.2 Amplitude Damping Ambiguity

Theorem 4 (Amplitude Damping Collapse)**.** *For the amplitude damping channel $\mathcal{A}_\gamma$:*

1. $\mathcal{A}_\gamma(|0\rangle\langle 0|) = \{|0\rangle\langle 0|\}$ *(unique)*
2. $\mathcal{A}_\gamma(|1\rangle\langle 1|) = \left\{\alpha|0\rangle\langle 0| + \beta|1\rangle\langle 1| : \alpha = 1 - \frac{1-\gamma}{\beta}, \beta \in [\gamma, 1]\right\}$
3. *The ground state $|0\rangle$ is uniquely identifiable, but excited states are not*

5.3 Fiber Noise Channel Ambiguity

For combined amplitude and phase damping $\mathcal{F}_{\gamma,\lambda}$:

$$\mathcal{A}(\sigma) = \left\{\rho : \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix} \to \begin{pmatrix} 1-(1-\gamma)\rho_{11} & \sqrt{1-\lambda}(1-\gamma)^{1/2}\rho_{01} \\ \sqrt{1-\lambda}(1-\gamma)^{1/2}\rho_{10} & (1-\gamma)\rho_{11} \end{pmatrix} = \sigma\right\} \tag{1}$$

The ambiguity increases with both γ and λ.

6 Information-Theoretic Characterization

6.1 Holevo Bound and Ambiguity

Theorem 5 (Holevo Bound Ambiguity Relation)**.** *The maximum accessible information is bounded by:*

$$\chi(\{p_x, \rho_x\}) \leq S\left(\sum_x p_x \mathcal{N}(\rho_x)\right) - \sum_x p_x S(\mathcal{N}(\rho_x))$$

The ambiguity manifests as the difference:

$$\mathcal{A}_{info} = I_{acc} - I_{max} \geq 0$$

where I_{acc} is accessible information and I_{max} is mutual information.

6.2 Channel Capacity and Ambiguity

Theorem 6 (Capacity-Ambiguity Trade-off)**.** *For a quantum channel $\mathcal{N}$:*

$$C_\chi(\mathcal{N}) = \sup_{\{p_x,\rho_x\}} \left[S\left(\sum_x p_x \mathcal{N}(\rho_x)\right) - \sum_x p_x S(\mathcal{N}(\rho_x)) \right]$$

The optimal ensemble minimizes ambiguity while maximizing distinguishability.

7 Geometric Interpretation

7.1 Bloch Sphere Representation

For qubit channels, the ambiguity can be visualized on the Bloch sphere:

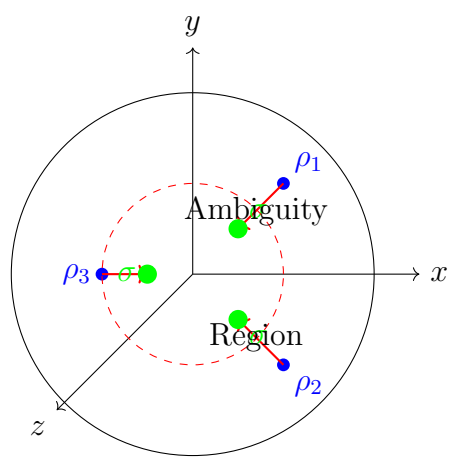

Figure 1: Geometric representation of ambiguity: Multiple input states (blue) map to the same output state (green) under the noisy channel (red arrows). The dashed circle represents the ambiguity region where inversion is non-unique.

7.2 Contraction Maps

Definition 6 (Channel Contraction)**.** *A quantum channel $\mathcal{N}$ is contracting if:*

$$\|\mathcal{N}(\rho_1) - \mathcal{N}(\rho_2)\|_1 \leq \eta \|\rho_1 - \rho_2\|_1$$

with $0 \leq \eta < 1$ *for some norm* $\|\cdot\|_1$.

Theorem 7. *Any noisy channel is contracting, with contraction factor* η *related to noise strength.*

8 Numerical Quantification of Ambiguity

8.1 Algorithm for Ambiguity Quantification

Require: Channel $\mathcal{N}$, input states $\{\rho_i\}_{i=1}^N$, tolerance ϵ
Ensure: Ambiguity matrix A, ambiguity measure $\mathcal{A}$

```
Initialize ambiguity matrix A_ij = 0 for i, j = 1, ..., N
for i = 1 to N do
    for j = i + 1 to N do
        Compute δ = ||N(ρ_i) − N(ρ_j)||_1
        if δ < ε then
            A_ij = A_ji = 1
        end if
    end for
end for
Compute ambiguity measure: A = (1/N^2) Σ_{i,j} A_ij
return A
```

9 Decoding Strategies Under Ambiguity

9.1 Maximum Likelihood Decoding

Given measurement outcome y, the ML decoder chooses:

$$\hat{x}_{\text{ML}} = \arg\max_{x \in \mathcal{X}} \mathbb{P}(y|x)$$

Ambiguity arises when multiple x have (nearly) equal likelihood.

9.2 Bayesian Decoding

With prior $p(x)$, the Bayesian decoder chooses:

$$\hat{x}_{\text{Bayes}} = \arg\max_{x \in \mathcal{X}} p(x|y) = \arg\max_{x \in \mathcal{X}} p(x)\mathbb{P}(y|x)$$

9.3 Minimum Error Probability

The optimal decoder minimizes:

$$P_e = \sum_x p(x) \sum_{y:\hat{x}(y)\neq x} \mathbb{P}(y|x)$$

Ambiguity increases P_e by creating decision boundaries where errors are unavoidable.

10 Capacity-Ambiguity Trade-off

Theorem 8 (Ambiguity-Capacity Relation)**.** *For a quantum channel $\mathcal{N}$, there exists a fundamental trade-off:*

$$C_\chi(\mathcal{N}) \leq \log_2\left(\frac{1}{\mathcal{A}_{\min}}\right)$$

where $\mathcal{A}_{\min}$ is the minimum achievable ambiguity for any encoding.

Proof. The accessible information is limited by the distinguishability of output states. Higher ambiguity reduces distinguishability, thus reducing capacity. □

11 Fiber Communication Specific Analysis

11.1 Practical Implications

For fiber-optic quantum communication:

1. **Photon Loss Ambiguity**: Cannot distinguish between weak signal and background noise
2. **Phase Ambiguity**: Cannot distinguish $|\psi\rangle$ from $e^{i\phi}|\psi\rangle$ without reference
3. **Polarization Ambiguity**: Fiber birefringence creates rotation ambiguities

11.2 Strategies to Reduce Ambiguity

1. **Encoding**: Use non-orthogonal states with better noise resilience
2. **Adaptive Measurements**: Adjust measurement basis based on channel estimation
3. **Error-Correcting Codes**: Introduce redundancy to resolve ambiguities
4. **Reference Signals**: Send known reference states to calibrate measurements

12 Formal Definition of Inversion Difficulty

Definition 7 (Inversion Difficulty)**.** *The inversion difficulty of channel $\mathcal{N}$ is quantified by:*

$$\mathcal{D}(\mathcal{N}) = \inf_{\mathcal{R} \in CPTP} \sup_{\rho \in \mathcal{D}(\mathcal{H})} \|\mathcal{R} \circ \mathcal{N}(\rho) - \rho\|_1$$

where the infimum is over all possible recovery operations.

Definition 8 (Decoding Complexity)**.** *The decoding complexity for channel $\mathcal{N}$ with encoding $\mathcal{E}$ is:*

$$\mathcal{C}_{dec}(\mathcal{N}, \mathcal{E}) = \min_{\mathcal{M}} \left[P_e(\mathcal{M}) + \alpha \cdot cost(\mathcal{M})\right]$$

where $\mathcal{M}$ ranges over all measurement strategies, P_e is error probability, and cost measures implementation complexity.

13 Mathematical Characterization

13.1 Condition Number Analysis

The inversion problem can be analyzed through the condition number of the channel:

Definition 9 (Channel Condition Number)**.** *For a quantum channel $\mathcal{N}$, define:*

$$\kappa(\mathcal{N}) = \sup_{\rho_1 \neq \rho_2} \frac{\|\rho_1 - \rho_2\|}{\|\mathcal{N}(\rho_1) - \mathcal{N}(\rho_2)\|}$$

Large κ indicates ill-conditioned inversion.

Theorem 9. *For noisy channels, $\kappa(\mathcal{N}) \to \infty$ as noise increases, making inversion arbitrarily ill-conditioned.*

13.2 Singular Value Analysis

Decompose the channel in terms of singular values:

$$\mathcal{N}(\rho) = \sum_i \sigma_i U_i \rho U_i^\dagger$$

Small singular values σ_i correspond to directions that are strongly attenuated by noise, creating inversion ambiguity.

14 Conclusion

1. **Fundamental Ambiguity**: Noisy quantum channels are inherently non-injective, creating unavoidable decoding ambiguity
2. **Quantifiable Measure**: Ambiguity can be quantified through ambiguity sets, contraction factors, and condition numbers
3. **Capacity Limitation**: Ambiguity directly limits achievable communication rates via the Holevo bound
4. **Practical Implications**: For fiber quantum communication, ambiguity manifests as photon loss confusion, phase uncertainty, and polarization rotation ambiguity
5. **Design Guidelines**: Communication strategies must account for ambiguity through appropriate encoding, measurement, and error correction

Appendix: Mathematical Proofs

Proof of Channel Non-Injectivity. Let $\mathcal{N}$ be a non-unitary CPTP map. Then $\mathcal{N}$ has Kraus representation $\mathcal{N}(\rho) = \sum_i E_i \rho E_i^\dagger$ with $\sum_i E_i^\dagger E_i = I$. For non-unitary channels, the operators $\{E_i\}$ are not all proportional to unitaries.

Consider two different purifications of the same mixed state. These pure states become identical after tracing out the environment, demonstrating non-injectivity.

More concretely, for amplitude damping:

$$\mathcal{A}_\gamma(|1\rangle\langle 1|) = \begin{pmatrix} \gamma & 0 \\ 0 & 1-\gamma \end{pmatrix}$$

Any state $\rho = \begin{pmatrix} a & b \\ b^* & 1-a \end{pmatrix}$ with $a = \gamma$ maps to the same output regardless of b, showing non-injectivity. □

Proof of Contraction Property. For any CPTP map $\mathcal{N}$:

$$\|\mathcal{N}(\rho_1) - \mathcal{N}(\rho_2)\|_1 \leq \|\rho_1 - \rho_2\|_1$$

by the contractivity of trace distance under CPTP maps. For noisy channels, the inequality is often strict, indicating contraction.

Specifically, for depolarizing channel:

$$\|\mathcal{D}_p(\rho_1) - \mathcal{D}_p(\rho_2)\|_1 = (1-p)\|\rho_1 - \rho_2\|_1$$

showing contraction factor $\eta = 1 - p < 1$ for $p > 0$. □